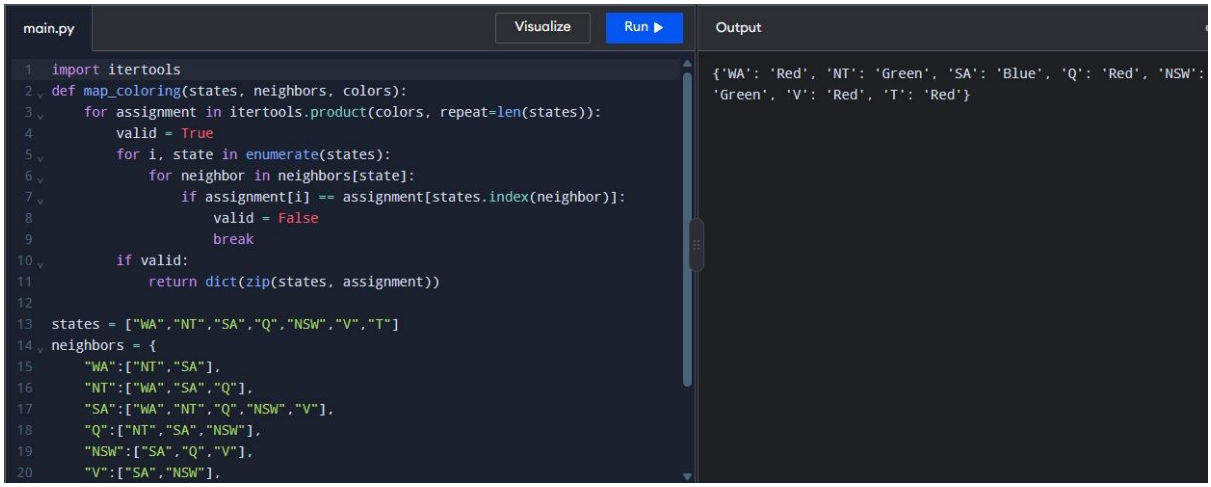
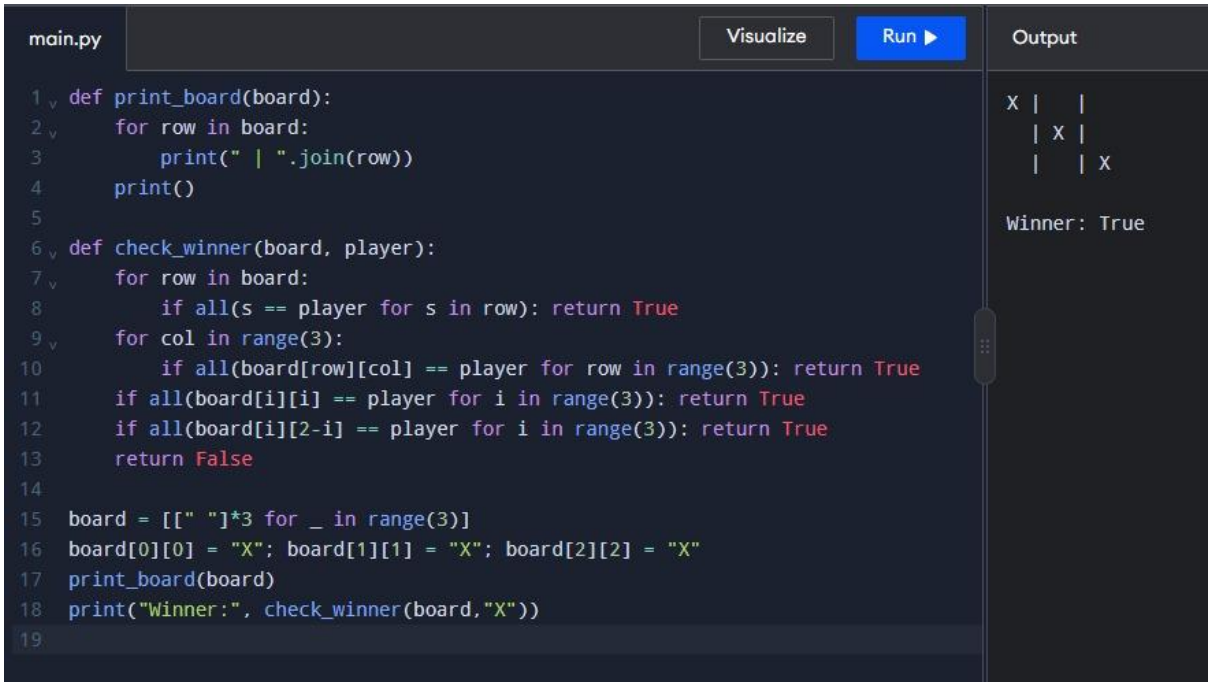
CSA1712 ARTIFICIAL INTELLIGENCE

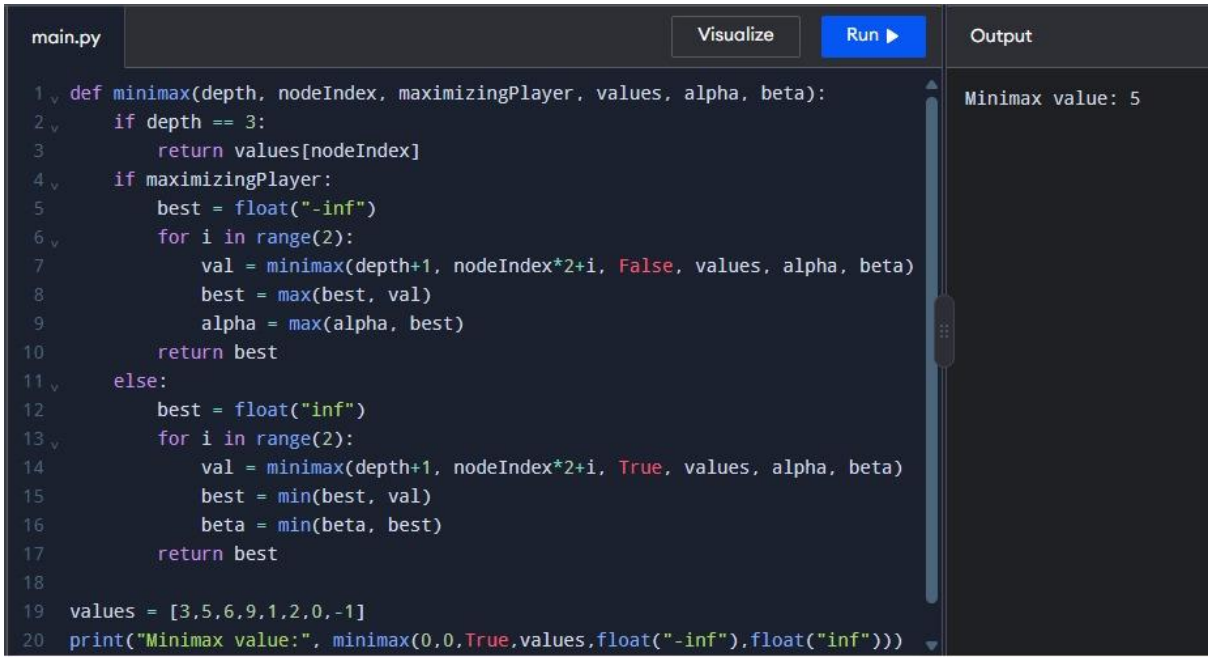
EXP 11: CSP



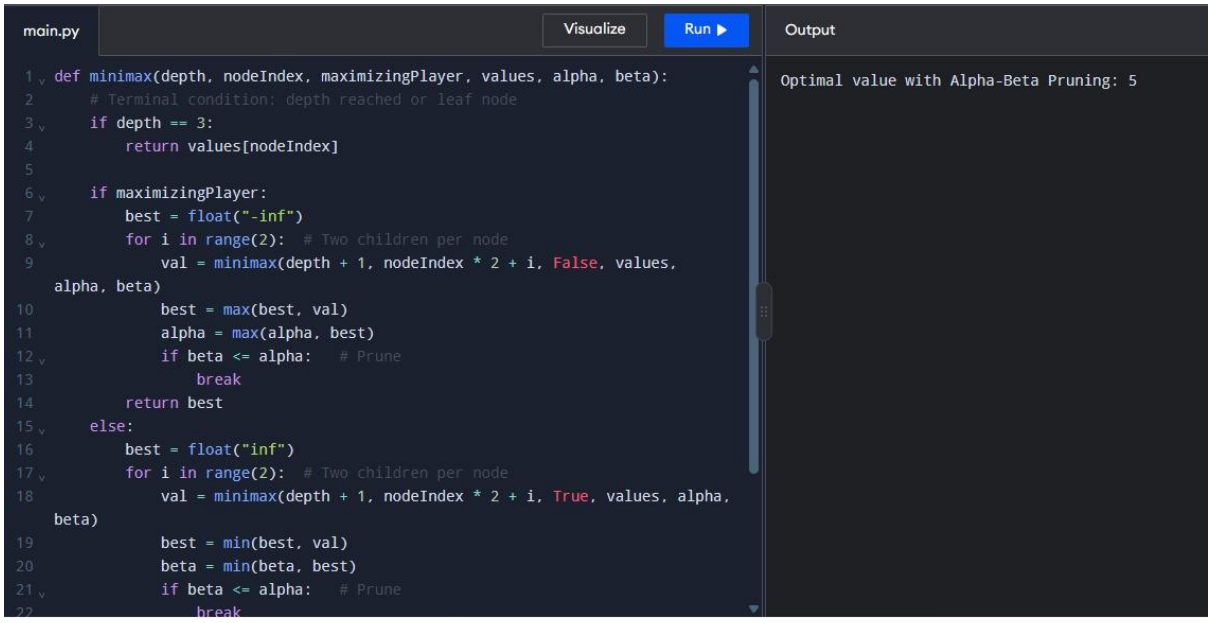
EXP 12: TIC TAC TOE



EXP 13: MINMAX



EXP 14: ALPHA & BETA



EXP 15: DECISION TREE

main.py

Visualize

Run ▶

```
class Node:
    def __init__(self, feature=None, threshold=None, left=None, right=None,
value=None):
        self.feature = feature      # Feature index
        self.threshold = threshold  # Split value
        self.left = left            # Left child
        self.right = right          # Right child
        self.value = value          # Class label if leaf

def build_tree(X, y, depth=0, max_depth=2):
    # If all labels are the same, return a leaf
    if len(set(y)) == 1:
        return Node(value=y[0])
    # Stop if max depth reached
    if depth >= max_depth:
        return Node(value=max(set(y), key=y.count))

    # Split on first feature for simplicity
    feature = 0
    threshold = 0.5
    left_X, left_y, right_X, right_y = [], [], [], []
    for xi, yi in zip(X, y):
        if xi[feature] <= threshold:
            left_X.append(xi)
```

Output

```
Prediction for Sunny (0): No
Prediction for Rainy (1): Yes
```